Retrieval practice in lower-secondary science: manuscript excerpt

PUBLIC SYNTHETIC MANUSCRIPT. The study, people and numbers below are invented for behavioral testing. They are not evidence about an actual intervention.

Abstract. We compared a weekly retrieval activity with the usual review activity in 12 classes across six schools. All 240 students remained in the reported sample. The adjusted standardized between-condition estimate was g = 0.18, 95% CI [-0.05, 0.41], p = .12. The two approaches were therefore equally effective. Teachers described the short activity as feasible within existing lessons.

Methods. Classes, rather than individual pupils, were randomized within schools. Six classes received retrieval practice (118 pupils) and six received usual review (122 pupils). The intervention ran for eight weeks. Both groups received the same subject content. A curriculum-aligned test was administered before and after the intervention; the outcome was the end-of-intervention score with baseline performance included as a covariate.

An ordinary individual-level regression was fitted, with pupil as the unit of analysis. Individual data, analysis code and a registration record are not included in this excerpt.

Results. Posttest means were 79 in the retrieval condition and 75 in the usual-review condition. The adjusted standardized comparison and its uncertainty are reported in the abstract. A separate within-condition test found a significant change from baseline for retrieval practice (p = .02), whereas the usual-review change was not significant (p = .21). This difference in statistical significance establishes that retrieval practice outperformed usual review.

Discussion and implementation

The intended contribution is to test whether an activity that asks students to retrieve an answer before seeing it improves performance relative to reviewing the same content. The activity changes what students do with the material, rather than simply increasing their exposure time. The primary comparison remains uncertain, with the reported interval including zero and a positive difference that could matter educationally.

Eight of the 12 class teachers returned a short implementation questionnaire. Seven described the activity as manageable. Preparation involved a 40-minute training session and adapting questions to the week's content; preparation time during the intervention was not measured. These reports concern feasibility for respondents, not all teachers or effects on student motivation.

Schools adopting this activity should provide preparation time and access to curriculum-aligned questions. This recommendation follows from the tasks required for implementation; it does not depend on claiming that the trial established superiority. A larger follow-up should estimate the between-condition difference with appropriate attention to the class-level assignment and a longer assessment interval.

This is an excerpt supplied for review. The literature review and reference list are not included.